

B2W Management Reporting API Help Guide

How to reach your Management Reporting (MR) API, sign in, send the right headers, and pull report data from the Data Warehouse. Paste your Ops URL into the card to fill in every example.

Understand the basics

What the MR API covers, where its data comes from, and where to find it.

1 What is the Management Reporting API?

SHORT ANSWER

The MR API is a read-only REST API over the B2W Management Reporting Data Warehouse. It returns report data such as estimates, pay items, tasks, cost components, bid results, change orders, and vendor and subcontractor quotes. Every operation is a GET.

The MR API v2 catalog lists 42 functional areas and 106 operations: a data call and a `/schema` call for each area, plus Login and Ping.

SOURCE

Estimate databases

Where estimators do their work.



DATA WAREHOUSE

Management Reporting

Loaded from the source databases.



MR API

Read-only report data

GET requests only.



Report data, not live estimates

Records come from the Data Warehouse, not straight from Estimate. Each record shows where it came from (`SourceDatabaseName`) and when (`LastRetrievedFromSourceDatabaseOn`). If the API shows old numbers, check those fields first. To change data, use Estimate or the Estimate API.

2

How do I find my MR API URL?

SHORT ANSWER

Use the same formula with the `MRAPI_` prefix:

`https://{cluster}/MRAPI_{environment}` . For `<environment>`, that is
`https://<cluster>.b2w.trimble.com/MRAPI_<environment>` .

`https://<cluster>.b2w.trimble.com/MRAPI_<environment>`

CLUSTER (HOST) PREFIX ENVIRONMENT

Check an environment

TOOL

Environment checker

Builds all three API addresses and checks each one

Paste the Ops URL to build all three API addresses and check each one.

3

Where is the MR API documentation?

SHORT ANSWER

Add `/doc/index.html` to the MR API URL. Like the Estimate API, it is a standard Swagger page with setup notes at the top and every functional area below.

https://<cluster>.b2w.trimble.com/MRAPI_<environment>/doc/index.html

B2W Management Reporting API v2 OAS3

General Information (click to expand/collapse)

Post-Installation Step (click to expand/collapse)

Configuration (click to expand/collapse)

Usage Guidelines (click to expand/collapse) ①

Testing Guidelines (click to expand/collapse) ②

Troubleshooting Guidelines (click to expand/collapse)

API Documentation

The following sections provide detailed information regarding every functional area where data is retrieved from the Management Reporting Data Warehouse Database. Click on the section title to expand/collapse that section.

BidResultsCompetitor ③



BidResultsCompetitorItemPrice



Figure 1. ① Usage Guidelines covers login and Client ID Security. ② Testing Guidelines links the official Postman collection. ③ Functional areas, each read from the Data Warehouse.

The machine-readable catalog is at `/doc/v2/MRAPI.json`.

PART 2

Sign in

Get a token, then send it with the headers every MR call needs.

4 How do I log in and get a Bearer Token?

SHORT ANSWER

Exactly like the Estimate API: send `GET {baseUrl}/Login` with `userName` (DOMAIN\user) and `password` headers for an Active Directory account with access to Management Reporting. The `AccessToken` in the response is your bearer token.

POSTMAN

GET `{{BaseURL}}/Login`

HEADERS

userName `{{UserName}}` (DOMAIN\user)

password `{{Password}}`

POWERSHELL

POWERSHELL

```
$baseUrl = "https://<cluster>.b2w.trimble.com/MRAPI_<environment>"

$login = Invoke-RestMethod -Method Get -Uri "$baseUrl/Login" -Headers @{
    userName = "DOMAIN\jsmith"
    password = "<password>"
}

$token = $login.AccessToken
```

CURL

CURL · BASH, MACOS, GIT BASH

```
curl -s 'https://<cluster>.b2w.trimble.com/MRAPI_<environment>/Login' \
-H 'userName: DOMAIN\jsmith' \
-H 'password: <password>'
```

RAW HTTP

HTTP REQUEST

```
GET /MRAPI_<environment>/Login HTTP/1.1
Host: <cluster>.b2w.trimble.com
userName: DOMAIN\jsmith
password: <password>
```

The responses match the Estimate API: **200** with the AccessToken, **400** when a header is missing (the API answers "Both userName and password must be specified in the request header"), **401** when the user is not authorized, and **500** for server errors.

Is a token expired, or whose is it? Paste it into the [token inspector](#). It shows when the token was issued and expires, and who it belongs to. It decodes the token on this page only and clears it when you close the window.

5 Which headers does every call need?

SHORT ANSWER

Every data call needs `Authorization: Bearer <AccessToken>` and a `DatabaseName` header naming the Data Warehouse database. Add `ClientID` and `ClientSecret` when Client ID Security is on, and `EstimateREF` on the eight calls that return one estimate's items.

HEADER	WHEN	VALUE
<code>Authorization</code>	Every data call	<code>Bearer <AccessToken></code> from Login

HEADER	WHEN	VALUE
DatabaseName	Every data call	The Management Reporting Data Warehouse database. The official collection's placeholder is B2WDataWarehouse .
ClientID · ClientSecret	When Client ID Security is on	The values set in the API; a mismatch returns 401
EstimateREF	The eight estimate-specific calls below	The ObjectID of the estimate

Calls that need EstimateREF

FOR ONE ESTIMATE (SEND ESTIMATEREF)	ACROSS ALL ESTIMATES (NO ESTIMATEREF)
/Estimate/PayItem · /Estimate/PayItemDetail	/Estimate/PayItemAll
/Estimate/IndirectItem · /Estimate/IndirectItemDetail	/Estimate/IndirectItemAll
/Estimate/Task · /Estimate/TaskDetail	/Estimate/TaskAll
/Estimate/WBSLevel · /Estimate/WBSLevelDetail	/Estimate/WBSLevelAll

The `Detail` versions return the same items with full information. Get an estimate's `ObjectID` from `GET /Estimate` first.

6

How do I set up the MR Postman collection?

SHORT ANSWER

Download the official MRAPI collection from the docs, import it, and fill in its variables. Like the Estimate collection, its pre-request script logs in for you before every request.

1 Download the collection

Click [Download the MR collection](#). It is the official collection from the API docs, also linked under **Testing Guidelines**. It has 107 requests.

2 Import it

Press `Ctrl + O` in Postman and drop the file in.

3 Fill in the variables

Set the values below on the collection's **Variables** tab and save.

VARIABLE	WHAT GOES IN IT	EXAMPLE
BaseUrl	The MR API URL (the file ships with <code>http://localhost/MRAPI</code>)	<code>https://<cluster>.b2w.trimble.com/MRAPI_<environment></code>
DatabaseName	The Data Warehouse database	<code>B2WDataWarehouse</code> (placeholder)
UserName · Password	Active Directory account, as <code>domain\username</code>	Your own values
ClientID · ClientSecret	Only when Client ID Security is on	Leave blank otherwise
EstimateREF	An estimate's ObjectID, for the estimate-specific requests	<code>DA45D31C-05FC-4C64-B1C8-8CE3E8E69D23</code> (sample)
Token	Set automatically by the pre-request script	Leave it

Make calls

Pull report data, page through it, and find the right endpoint.

7 How do I read, filter, and page report data?

SHORT ANSWER

Every GET returns an `Items` array and a `Pagination` block that tells you how many records exist. Use OData to filter and page; `contains()` works for text searches.

POSTMAN

GET

`{{BaseURL}}/Estimate/PayItem?$filter=contains(Description, 'Mobilization')`

HEADERS

DatabaseName `{{DatabaseName}}`

EstimateREF `{{EstimateREF}}`

AUTHORIZATION

Bearer Token `{{Token}}`, set by the pre-request script

POWERSHELL

POWERSHELL · CONTINUES FROM QUESTION 4

```
$headers = @{
    Authorization = "Bearer $token"
    DatabaseName  = "<Data Warehouse database name>"
    EstimateREF   = "<ObjectID of the estimate>"
}

$result = Invoke-RestMethod -Uri "$baseUrl/Estimate/PayItem" -Headers $headers -Body
@{ '$filter' = "contains(Description, 'Mobilization')" }
$result.Items
$result.Pagination # CurrentPage, ItemsOnPage, PageSize, TotalItems
```

CURL

CURL · BASH, MACOS, GIT BASH

```
curl -s -G 'https://<cluster>.b2w.trimble.com/MRAPI_<environment>/Estimate/PayItem' \
  --data-urlencode "\$filter=contains(Description, 'Mobilization')" \
  -H 'Authorization: Bearer <AccessToken>' \
  -H 'DatabaseName: <Data Warehouse database name>' \
  -H 'EstimateREF: <ObjectID of the estimate>'
```

RAW HTTP

HTTP REQUEST

```
GET /MRAPI_<environment>/Estimate/PayItem?$filter=contains(Description,
'Mobilization') HTTP/1.1
Host: <cluster>.b2w.trimble.com
Authorization: Bearer <AccessToken>
DatabaseName: <Data Warehouse database name>
EstimateREF: <ObjectID of the estimate>
```

The [filter helper](#) in the request builder lists each endpoint's fields and writes the `$filter` for you, with text quoted, IDs left unquoted, and dates in the format the API expects.

RESPONSE · GET /ESTIMATE (SHORTENED CATALOG EXAMPLE)

```
{
  "Items": [
    {
      "Title": "Fox Hill Development",
      "ObjectID": "02e2bd02-b667-4478-a1c6-76dedfb8958f",
      "EstimateNumber": "0012",
      "BidNumber": "64687",
      "EstimateStatus": "Pending",
      "SourceDatabaseName": "B2WSample",
      "LastRetrievedFromSourceDatabaseOn": "2025-12-07T23:40:17.823",
      "TotalBidPriceWithTax": 12627661.59
    }
  ],
  "Pagination": {
    "CurrentPage": ".../Estimate?$orderBy=Title&$skip=0&$top=5",
    "ItemsOnPage": 1,
    "PageSize": 5,
    "TotalItems": 1
  }
}
```



Use TotalItems to page

Pagination.TotalItems tells you how many records match. Keep adding \$skip in steps of your \$top (100 at most) until you have them all.

8

Which endpoints are available?

SHORT ANSWER

The MR API v2 catalog lists 106 operations, all GET. Each area has a data call and a /schema call. Search below, then check Swagger for the fields.

TOOL Endpoint explorer

Showing 54 of 54 endpoint groups · 106 operations in all

/Login

GET

userName + password headers → AccessToken

/Ping/{input}

GET

No login needed

/Estimate

GET

+ /schema

/Estimate/PayItem

GET

+ /schema **EstimateREF**

/Estimate/PayItemDetail

GET

+ /schema **EstimateREF**

/Estimate/PayItemAll

GET

+ /schema

/Estimate/IndirectItem

GET

+ /schema **EstimateREF**

/Estimate/IndirectItemDetail

GET

+ /schema **EstimateREF**

/Estimate/IndirectItemAll

GET

+ /schema

/Estimate/Task

GET

+ /schema **EstimateREF**

/Estimate/TaskDetail

GET

+ /schema **EstimateREF**

/Estimate/TaskAll

GET

+ /schema

/Estimate/WBSLevel

GET

+ /schema **EstimateREF**

/Estimate/WBSLevelDetail

GET

+ /schema **EstimateREF**

/Estimate/WBSLevelAll

GET

+ /schema

/BidResultsCompetitor

GET

+ /schema

/BidResultsCompetitorItemPrice

GET

+ /schema

/BidResultsItem

GET

+ /schema

/BidResultsSettings

GET

+ /schema

/BidSchedule

GET

+ /schema

/ItemBidClosingInformation

GET

+ /schema

/MinorityParticipationSummary

GET

+ /schema

/CrewComponent

GET

+ /schema

/EquipmentComponent

GET

+ /schema

/LaborComponent

GET + /schema

/MiscellaneousComponent

GET + /schema

/EquipmentBurdenInstance

GET + /schema

/TruckingTimeBreakdownInstance

GET + /schema

/ItemTaskQuickCostCategory

GET + /schema

/CostComponentTag

GET + /schema

/Competitor

GET + /schema

/Subcontractor

GET + /schema

/MaterialManufacturer

GET + /schema

/RFQGroupVendor

GET + /schema

/VendorRFQGroup

GET + /schema

/ChangeOrder

GET + /schema

/MaterialComponent

GET + /schema

/TruckingComponent

GET + /schema

/LaborBurdenInstance

GET + /schema

/ItemTaskComponent

GET + /schema

/ItemTaskSubcontractorQuote

GET + /schema

/Customer

GET + /schema

/Vendor

GET + /schema

/Material

GET + /schema

/VendorQuote

GET + /schema

/RFQGroupSubcontractor

GET + /schema

/SubcontractorRFQGroup

GET + /schema

/ChangeOrderLog

GET + /schema

/DataWarehouseInformation

GET + /schema

/SourceDatabase

GET + /schema

/CompatibleSourceDatabaseVersion

GET + /schema

/GlobalOptions

GET + /schema

/CategorizedValueInformation

GET + /schema

/UserDefinedField

GET + /schema

GET read /schema field definitions **EstimateREF** needs that header

Each card's **Build** button opens that endpoint in the [request builder](#), which adds the `DatabaseName` and `EstimateREF` headers for you and writes the request for Postman, PowerShell, cURL, or raw HTTP.

Troubleshoot

The basic tests from the MR API docs, and fixes for the errors you will see most.

9 What does this error mean?

SHORT ANSWER

Paste the error into the decoder: a status line, a response body, a Postman or PowerShell error, or the request URL. It names the likely cause and links to the fix.

It knows the MR API's messages and SQL Server errors such as "Login failed for user 'NT AUTHORITY\NETWORK SERVICE'". Paste a successful response and it reads the `Pagination` block and `LastRetrievedFromSourceDatabaseOn` too, which answers "why is data missing?" and "why are the numbers old?". Decoding runs entirely in your browser.

10 How do I troubleshoot the MR API?

SHORT ANSWER

Run the app-up test with `/Ping/hello`, then the login test. Don't go further until both pass. Then check the headers, especially `DatabaseName` and `EstimateREF`.

01

App-up test

`GET /Ping/hello` should return 200 and `(server) hello`. If not, check that IIS and the **MRAPI** application pool are running and that the URL is correct. The environment checker in question 2 runs this for you.

02

Login test

`GET /Login` with `userName` and `password` headers should return an `AccessToken`.

03

Data call headers

Check `Authorization: Bearer` , `DatabaseName` , the client values if Client ID Security is on, and `EstimateREF` on estimate-specific calls.

Common errors

401 Data call: “The user is not authorized for this API.”

LIKELY CAUSE

The bearer token is missing, invalid, or expired, or the ClientID or ClientSecret is wrong when Client ID Security is on.

FIX

Log in again for a fresh token, then check the client headers.

Wrong data Empty results or the wrong company’s data

LIKELY CAUSE

The `DatabaseName` header is missing or names the wrong Data Warehouse database.

FIX

Confirm the Data Warehouse database name and resend.

No items Pay items, tasks, or WBS levels come back empty

LIKELY CAUSE

The `EstimateREF` header is missing or isn’t an estimate in this Data Warehouse.

FIX

Get the estimate’s ObjectID from `GET /Estimate` , or use the `...All` version of the call.

Old data “The API shows old numbers”

LIKELY CAUSE

The Data Warehouse hasn't been reloaded from the source Estimate database since the change.

FIX

Check `LastRetrievedFromSourceDatabaseOn` on the record, and the Data Warehouse load for that source database.

SQL “Login failed for user ‘NT AUTHORITY\NETWORK SERVICE’”

LIKELY CAUSE

The API's Network Service account lacks db_owner access to the Data Warehouse database, or the API's SQL Server connection string is wrong.

FIX

On-premises: grant the access and verify the connection string (the docs' Post-Installation Step).
Cloud: [contact B2W Support](#).



Still stuck? Contact B2W Support

Email support_b2w@trimble.com or call [+1 \(888\) 390-8822](tel:+18883908822). Include the environment name, the endpoint and method, the status code and response with secrets removed, and the date, time, and time zone. The [error decoder](#)'s **Copy a summary** button writes most of it for you.



B2W Management Reporting API Help Guide

Guides and tools for the B2W Ops, Estimate, and Management Reporting APIs, for customers, partners, and B2W Support. They draw on the APIs' own documentation and official Postman collections; examples were checked on October 2, 2026. Your environment's own API docs are always the source of truth. Credentials and tokens in screenshots are hidden.

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